



Depression brain circuit

A brain circuit with special light-sensing cells that could explain 'winter blues' or seasonal depression. <https://digit.in/jan19-69>



Wall-scaling robot

HAMR-E is a tiny robot with sticky feet, that can walk on vertical surfaces and even upside down. <https://digit.in/jan19-70>

Renne published a paper showing that most of the major mass extinction events are associated with increased volcanic activity. For example, the massive volcanic outpourings from the Siberian Traps are believed to be the cause of the Permian–Triassic (P–T) extinction event. It is the only known mass extinction event for insects, and about 90 percent of all life forms were wiped from existence. Life on Earth took ten million years to recover from the P–T extinction event. Courtillot and Renne proposed that increased volcanism was the general explanation for all mass extinction events.

While all the major extinction events are chronologically linked to known sources of increased volcanism, only the K–Pg extinction event is linked to an impact crater. Considering the rarity of the impact craters of a similar size, and the frequency of flood basalt plains, the authors show that periods of increased volcanism are more common than large asteroid strikes. The paper goes on to demonstrate that while the volcanism is sufficient to cause the extinction on its own, and the only clear evidence of an asteroid that could have caused a mass extinction is during a period with increased volcanism.

One of the biggest opponents of the Alvares hypothesis is Gerta Keller, a palaeontologist from Princeton University. Keller, along with Ashok Sahni from Punjab University and Sunil Bajpai from IIT-Roorkee have published research linking the massive volcanic outpourings in the Deccan traps to the K–Pg extinction event.

The igneous rock that we read about in the boring geography textbooks are actually the remains of massive floods of lava. The Deccan traps, an igneous province that covers portions of Gujarat, Madhya Pradesh, Maharashtra and Northern Karnataka, is one of the largest volcanic features in the world. The massive eruptions reached over 1000 km across India, and could well be the longest lava flows known to man. What remains of these layers of cooled down lava are

the stratified rocks that we see in the Deccan traps today.

According to the volcanism theory, the massive eruptions pumped sulfur dioxide and carbon dioxide into the atmosphere, over a period of a million years. The sulfur dioxide would have cooled the atmosphere, while the carbon dioxide would have heated it.



The Deccan Traps

The geological record shows mixed signals, with some rocks showing a period of cooling, while others showing a period of warming. One explanation is a period of cooling followed by a sudden period of warming, and a runaway greenhouse gas effect. The carbon cycle of the Earth was disturbed, and this led to the K–Pg extinction event. The volcanism also explains how the iridium got released into the atmosphere of the Earth, and was deposited all over it.

THE ONE-TWO PUNCH

While the debate got murky at times, the two theories are not totally at odds with each other. Researchers have been trying to accurately date the Chicxulub impactor, and some findings suggested that the asteroid had struck slightly before the extinction event. The best dating techniques now show that the strike coincided almost exactly with the extinction. However, that does not mean that the Chicxulub event was the only cause for the extinction of the dinosaurs. It could be that the asteroid

strike was the "last straw". The fossil record shows that almost all the non-avian dinosaurs had been wiped out before the asteroid struck, which would make the volcanism in the Deccan traps the only available explanation.

Even in the 2003 paper, Courtillot and Renne did not consider the Chicxulub impact as opposed to the volcanism

hypothesis. Instead, they proposed that the perceived consequences of the strike was lesser than previously thought. The meteor was not the single reason that the dinosaurs died, but it did manage to wipe out species that were already stressed. The authors maintain that volcanism was the main agent for the K–Pg extinction event.

There are some questions still to be answered in the one-two punch, or hybrid theory of dinosaur extinction. Essentially, instead of just a single factor, it considers both the volcanism and the impactor as contributing factors. Out of the two, the volcanism could have finished off the dinosaurs on their own. The question now arises on how much the volcanism and the strike contributed to the extinction. While the volcanism was spread out over a million years, about 90 percent of the lava flows are suspected to have occurred right around the K–Pg boundary. A more precise dating of the layers in the Deccan Traps is needed to resolve the remaining questions. 